

Midterm Test

1. Prove that if the series $\sum_{n=1}^{\infty} x_n$ is convergent, then $\lim_{n \rightarrow \infty} x_n = 0$. **1p**
2. Find the value of $\lim_{n \rightarrow \infty} \frac{1}{3n+1} + \frac{1}{3n+2} + \dots + \frac{1}{4n}$. **1p**
3. Study the convergence of the following: **3p**
 - (a) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.
 - (b) $\int_0^{\pi} \frac{1}{\sin^2 x} dx$.
 - (c) $\sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot \dots \cdot (3n-2)}{3^n n!}$.
4. Find the Taylor series around 0 and its radius of convergence for each of the following:
 - (a) $\ln(1-x)$. **1p**
 - (b) $\sqrt{1+x}$. **1p**
5. Consider the power series $\sum_{n=2}^{\infty} \frac{x^n}{n^2 - n}$.
 - (a) Find its radius of convergence and its convergence set. **1p**
 - (b) Find its sum for any x in the convergence set. **1p**